Supplementary Online Content

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This supplementary material has been provided by the authors to give readers additional information about their work.

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eMethods: Search Terms

(((“Congenital” [Subheading]
AND
(“Cytomegalovirus” [MeSH]
AND
(“Africa” [MeSH] OR “Asia” [MeSH] OR “Central America” [MeSH] OR “Developing Countries” [MeSH]
OR “Geographical Locations Category” [MeSH] OR “Latin America” [MeSH] OR
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Venezuela*[tiab] OR Viet nam*[tiab] OR Vietnam*[tiab] OR Vincent[tiab] OR West[tiab] OR West Bank[tiab] OR
Yemen*[tiab] OR Zambia*[tiab] OR Zimbabw*[tiab])
NOT
(“Animals”[MeSH]

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eTable 1. Study Level Characteristics

First Author	Publication year	Study Period	Country	ISO code	Screening Biological Specimen	Laboratory Method	Tested	Positive	Symptomatic (%)	Male (%)	Maternal age (Y)	Maternal Seroprevalence (%)	Study Quality (NOS)
Tsai	1996		Taiwan	TWN	Urine	Culture, PCR	1000	18		72		90	8
Sohn	1992	1989-1991	South Korea	KOR	Urine and cord blood	Culture	514	6			30	96	8
Yamada	2020	2009-2018	Japan	JPN	Urine	PCR	11736	56	41				7
Koyano	2011	2008-2010	Japan	JPN	Urine	PCR	21272	66	23		30		7
Numazaki	2004	1977-2002	Japan	JPN	Urine	Culture	11938	37	14	54			7
Yamaguchi	2017	2008-2015	Japan	JPN	Urine	PCR	23368	60	73				7
Endo	2009	2005-2007	Japan	JPN	Blood	PCR	1010	2	0	51	30		8
Uchida	2019	2009-2017	Japan	JPN	Urine	PCR	4125	9	11	52	30		8
Moteki	2018	2011-2016	Japan	JPN	DBS	PCR	9667	47	4	64			8
Torii	2019	2014-2017	Japan	JPN	Urine	PCR	685	11	0		31	67	9
Yamagishi	2006	2004-2005	Japan	JPN	Blood	PCR	1176	2	0				6
Zhang	2007	1997-2000	China	CHN	Urine	PCR	1159	71	24	54			7
Wang	2017	2011-2013	China	CHN	DBS, Saliva	PCR	10933	75	0			96	7
Putri	2019	2016-2017	Indonesia	IDN	Saliva, Urine	PCR	411	24	67			98	8
Dar	2008	2008	India	IND	Saliva, Urine	PCR	423	9	11		22	99	8
Viswanathan	2019	2015-2017	India	IND	Saliva, Urine, Blood	PCR	750	3	33	50	29	100	9
Sahiner	2015	2013-2014	Turkey	TUR	Saliva, Urine	PCR	944	18					6
Paradiz	2012	2007-2008	Slovenia	SVN	Urine, Blood	PCR, Culture	2841	4	0	50			8
Paixão	2009	2003-2004	Portugal	PRT	Blood	PCR	3600	38					6
Barlinn	2018	1999-2008	Norway	NOR	Blood	PCR	1349	3			30	54	9
Gayant	2005	1998-2000	Netherlands	NLD	Blood, Throat swab, Urine	Serology, PCR, Culture	7793	7	0			41	8
de Vries	2011	2007	Netherlands	NLD	Blood	PCR	6433	35					9
Barbi	1998	1994-1995	Italy	ITA	Saliva, Urine, Blood	PCR	1268	6	0			80	8
Barbi	2006	2002-2003	Italy	ITA	DBS	PCR	9032	16	13	53	32	80	9
Schlesinger	2003	1992-1993	Israel	ISR	Urine	PCR, Culture	2000	14	7	71		83	9
Barkai	2014	2011-2012	Israel	ISR	Saliva, Urine (for confirmation)	PCR, Culture	9845	47	21	67	31		8

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Barkai	2013	2009-2010	Israel	ISR	Infant cord blood, Urine	PCR, Culture	8105	22	14	50	30		8
Waters	2014	2011-2012	Ireland	IRL	Urine, Saliva	PCR	1044	2	0				8
Puhakka	2019	2012-2015	Finland	FIN	Saliva	PCR	19868	40	10		32		8
Arapovic	2020	2010-2019	Bosnia and Herzegovina	BIH	Blood, Saliva	PCR	1293	8	0		30	92	9
Foulon	2008	1996-2006	Belgium	BEL	Urine	Culture	14021	74	5			59	8
Halwachs-Baumann	2000	1993-1997	Austria	AUT	Urine, Blood	Serology, PCR, Culture	5967	13	23				9
Leruez-Ville	2017	2013-2015	France	FRA	Saliva	PCR	11715	44	20		32	61	8
Engman	2008	2003-2004	Sweden	SWE	Blood, Urine	PCR, Culture	6060	12	0				7
Karimian	2016	2014-2016	Iran	IRN	Urine	PCR	1617	8	38	57	27		9
Fahimzad	2013	2012	Iran	IRN	Saliva	PCR	620	2	0	53			9
Noorbakhsh	2020	2017	Iran	IRN	Blood	PCR	1174	4	0	57	28	95	8
Fowler	1993	1980-1990	USA	USA	Urine	Culture	27055	267	10				8
Boppana	1999	1991-1997	USA	USA	Urine, Saliva	DEAFF	20885	246	19				9
Kharrazi	2010	2004-2005	USA	USA	Blood	PCR	3972	28				79	8
Boppana	2010	2007-2008	USA	USA	DBS, Saliva	PCR, rapid Culture	20448	92		51	27		9
Pinninti	2016	2008-2012	USA	USA	Saliva, Urine	PCR, rapid Culture	73239	266					8
Boppana	2011	2008-2009	USA	USA	Saliva, Urine	PCR, rapid Culture	34989	177	0	51	27		9
Dollard	2021	2016-2019	USA	USA	Saliva, DBS, Urine	PCR	12554	56					9
Estripeaut	2007	2003-2004	Panama	PAN	Urine	PCR	317	2				84	7
Noyola	2003	2001	Mexico	MEX	Saliva	Culture	560	5				92	6
Luschsinger	1996	1989-1994	Chile	CHL	urine and saliva (throatswab)‡	Culture, PCR	658	12				98	4
Yamamoto	2001	1998-1999	Brazil	BRA	Urine, Blood	PCR, Culture	332	7	57			96	6
Cardoso	2015	2012	Brazil	BRA	Urine, Saliva	PCR	1000	13	8				8
Yamamoto	2011	2003-2009	Brazil	BRA	Urine, Saliva	PCR, culture	12195	121				96	9
Yamamoto	2020	2013-2017	Brazil	BRA	Urine, Saliva	PCR	11900	68	12	52	26		9
Rico	2021	2017-2018	Colombia	COL	Urine	PCR	711	6				98	7
Pathirana	2019	2016	South Africa	ZAF	Saliva, Urine	PCR	2685	66	18	48	28	100	8
Olusanya	2015	2012-2013	Nigeria	NGA	Saliva	PCR	263	10	20		30		9

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Madrid	2018	2014-2015	Mozambique	MOZ	Blood, Nasopharyngeal swab	PCR	117	7	14		22	100	8
van der Sande	2007	2002-2005	The Gambia	GMB	Urine	PCR	741	40	8	53		100	9
Otieno	2019	2015-2017	Kenya	KEN	Saliva, DBS	PCR	1078	39	3			93	8
Griffiths	1991	1983-1985	United Kingdom	GBR	Urine	Culture	2737	9	22			60	7
Stagno	1986	1978-1984	USA	USA	Urine	Culture	11124	82	4		27	54	8
Starr	1970	1967-1968	USA	USA	Urine	Culture	2147	26	19		23		6
Birnbaum	1969	1966-1967	USA	USA	Urine	Culture	545	3	67				5
Ahlfors	1984	1977-1982	Sweden	SWE	Urine	Culture	10328	50					6
Andersen	1979	1974-1977	Denmark	DNK	Urine	Culture	3060	12	33				7
Stagno	1977		USA	USA	Urine	Culture	939	23	13		22	82	8
Melish†	1973	1968-1970	USA	USA	Urine	Culture	1963	20	5				7
Stern	1973		United Kingdom	GBR	Urine	Culture	720	5	20				6
Peckham‡	1983	1979-1982	United Kingdom	GBR	Throat swab	Culture	14200	42	5			56	7
Larke	1980	1974-1975	Canada	CAN	Urine	Culture	15212	64	6				6
Schopfer	1978		Ivory Coast	CIV	Urine	Culture	2032	28	0			100	5
Stagno	1982		Chile	CHL	Urine	Culture	118	2	0		23	98	8
Yow	1988	1981-1986	USA	USA	Urine	Culture	3899	17	12		29	52	8
Embil	1975	1969-1971	Canada	CAN	Urine	Culture	542	3	33	56			7
Kamada	1983		Japan	JPN	Urine	Culture	2070	11	0			94	5
Hanshaw	1968	1967-1968	USA	USA	Urine	Culture	280	6	17				4
Montgomery	1980	1972-1975	USA	USA	Urine	Culture	954	9	78		20		8
Hildebrandt	1967	1964-1965	USA	USA	Urine	Culture	130	1	0			51	5
Leikikki	1978	1972	Finland	FIN	Urine	Culture	191	3				82	7

PCR: polymerase chain reaction; NOS: Newcastle-Ottawa Quality Assessment Scale; DBS: dry blood spot; DEAFF: early antigen fluorescent foci; ISO, International Organization for Standardization.

Footnote

†only urine screening results reported in the meta-analysis; ‡Throat swab proxy for saliva

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eTable 2: Clinical signs and symptoms of cCMV. Frequency with percentages of clinical signs of cCMV reported in the studies included in the meta-analysis.

Clinical signs and symptoms of cCMV	**Frequency (n/number of studies)**
jaundice	15/27 (56%)
elevated liver enzymes	3/27 (11%)
hepatosplenomegaly	13/27 (48%)
thrombocytopenia	16/27 (59%)
petechiae	9/27 (33%)
anemia	4/27 (15%)
seizures	4/27 (15%)
chorioretinitis	8/27 (30%)
sensorineural hearing loss (deafness)	16/27 (59%)
CNS involvement (microcephaly, intracranial calcifications, enlarged ventricles)	16/27 (59%)
sepsis	3/27 (11%)

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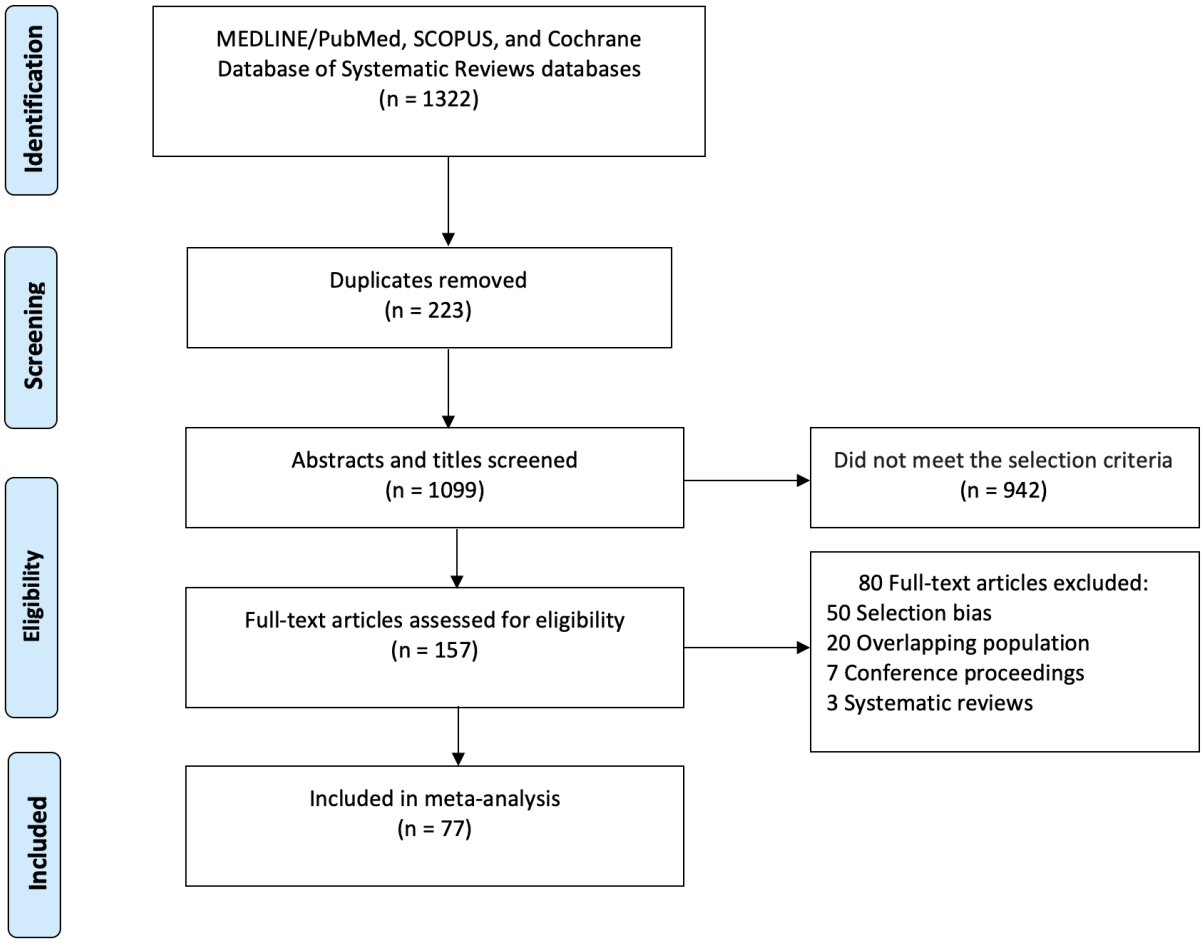
eFigure 1. PRISMA Diagram



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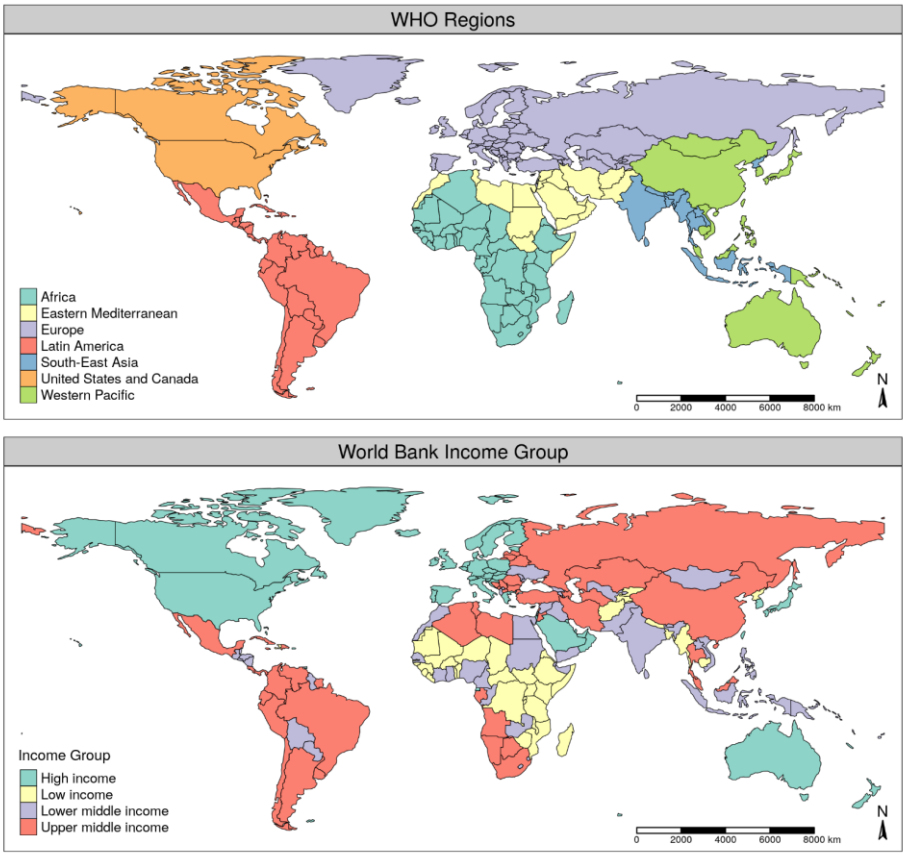

eFigure 2: Maps of WHO regions and income group. Maps were produced with the R software for statistical computing version 3.6.3. The world shapefile was retrieved from the **spData** package (version 0.3.0, https://CRAN.R-project.org/package=spData) and the maps created with the **tmap** package (version 3.2.2, https://CRAN.R-project.org/package=tmap).

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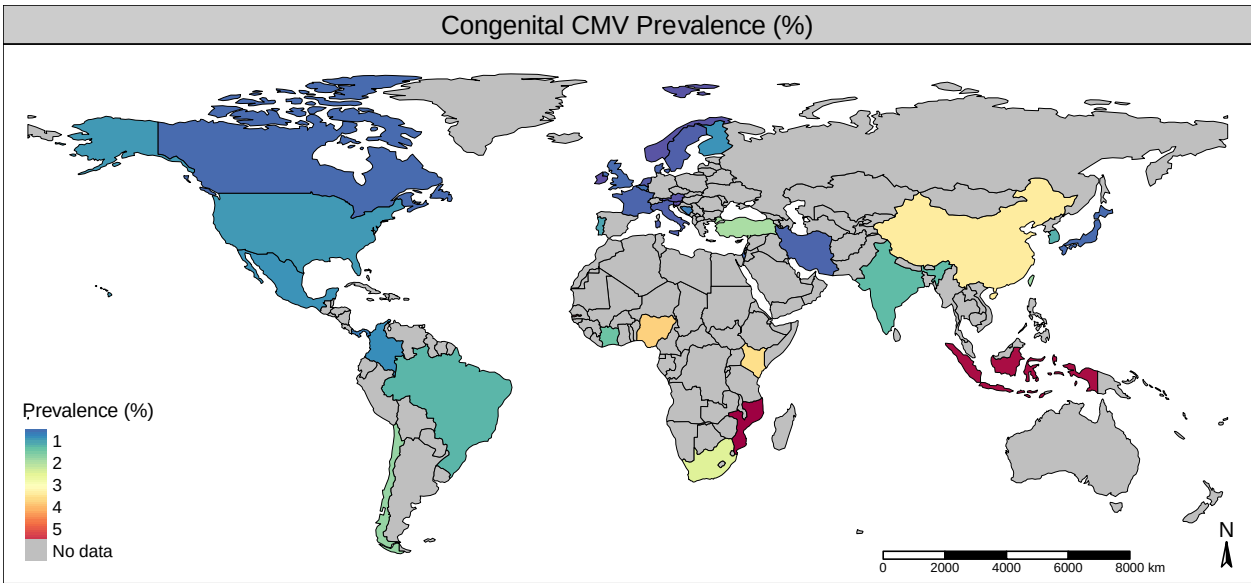

eFigure 3: Country-specific rates of cCMV in countries represented in the current meta-analysis. Birth prevalence of cCMV (%) by country. Countries with more than one study, the weighted rate was obtained using random effects models. Maps were produced with the R software for statistical computing version 3.6.3. The world shapefile was retrieved from the **spData** package (version 0.3.0, https://CRAN.R-project.org/package=spData) and the maps created with the **tmap** package (version 3.2.2, https://CRAN.R-project.org/package=tmap).

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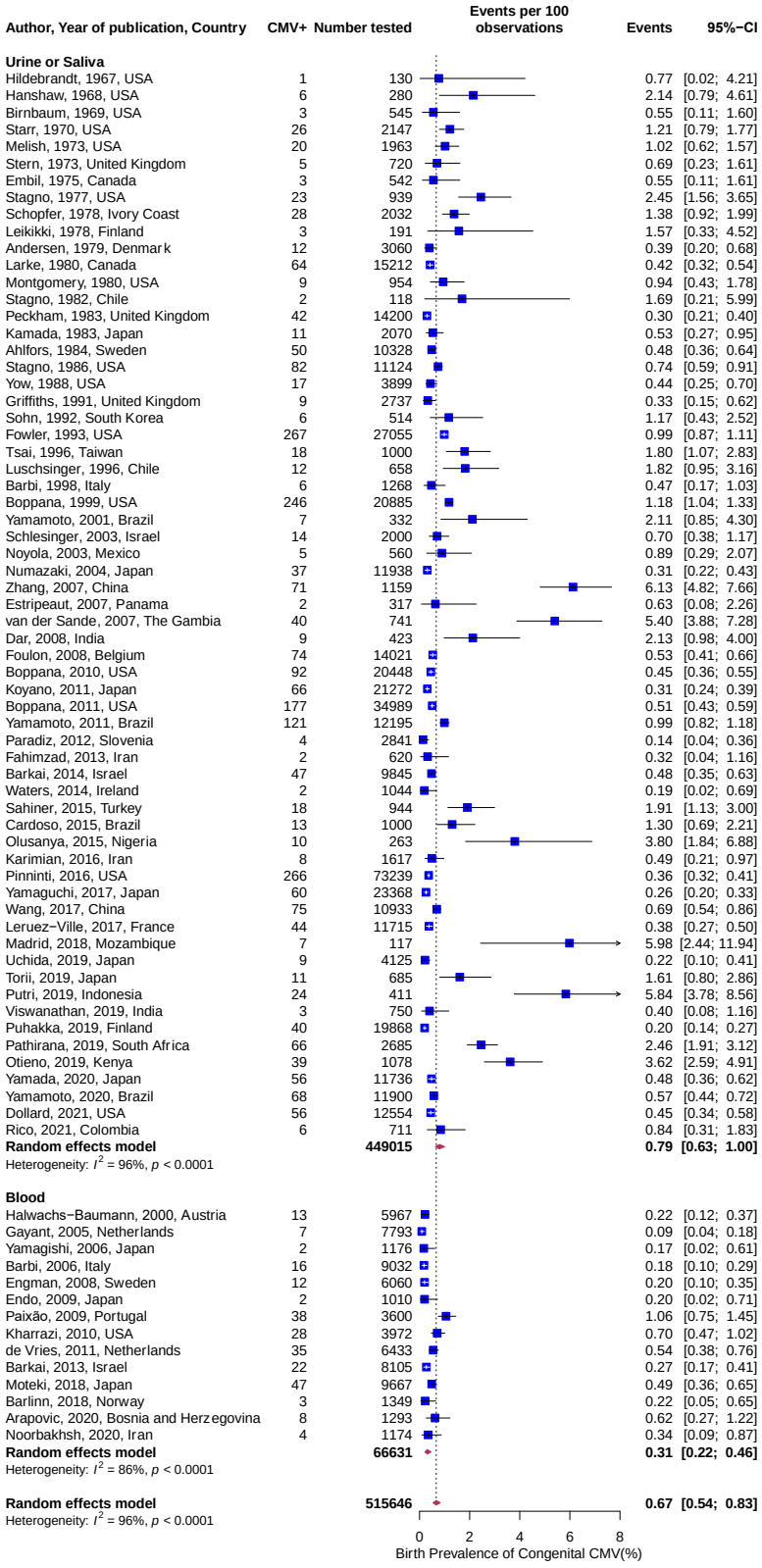

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eFigure 4: Screening methods (Blood vs. Urine or Saliva) for cCMV. Forest plot of cCMV rates by screen methods (Blood only Vs. Urine/Saliva). Effect size values represent congenital CMV cases expressed as a percentage and their corresponding 95% CI. Blue squares and their corresponding lines are the point estimates and 95% confidence intervals (95% CI). Maroon diamonds represent the pooled estimate of each subgroup's incidence (width denotes 95% CI). Heterogeneity by screening method: Urine or saliva *(I^2* = 96%, p for heterogeneity < 0.0001; 63 studies); HICs (I^2 = 86%, p for heterogeneity < 0.0001; 14 studies); p for comparing the subgroups <0.0001.

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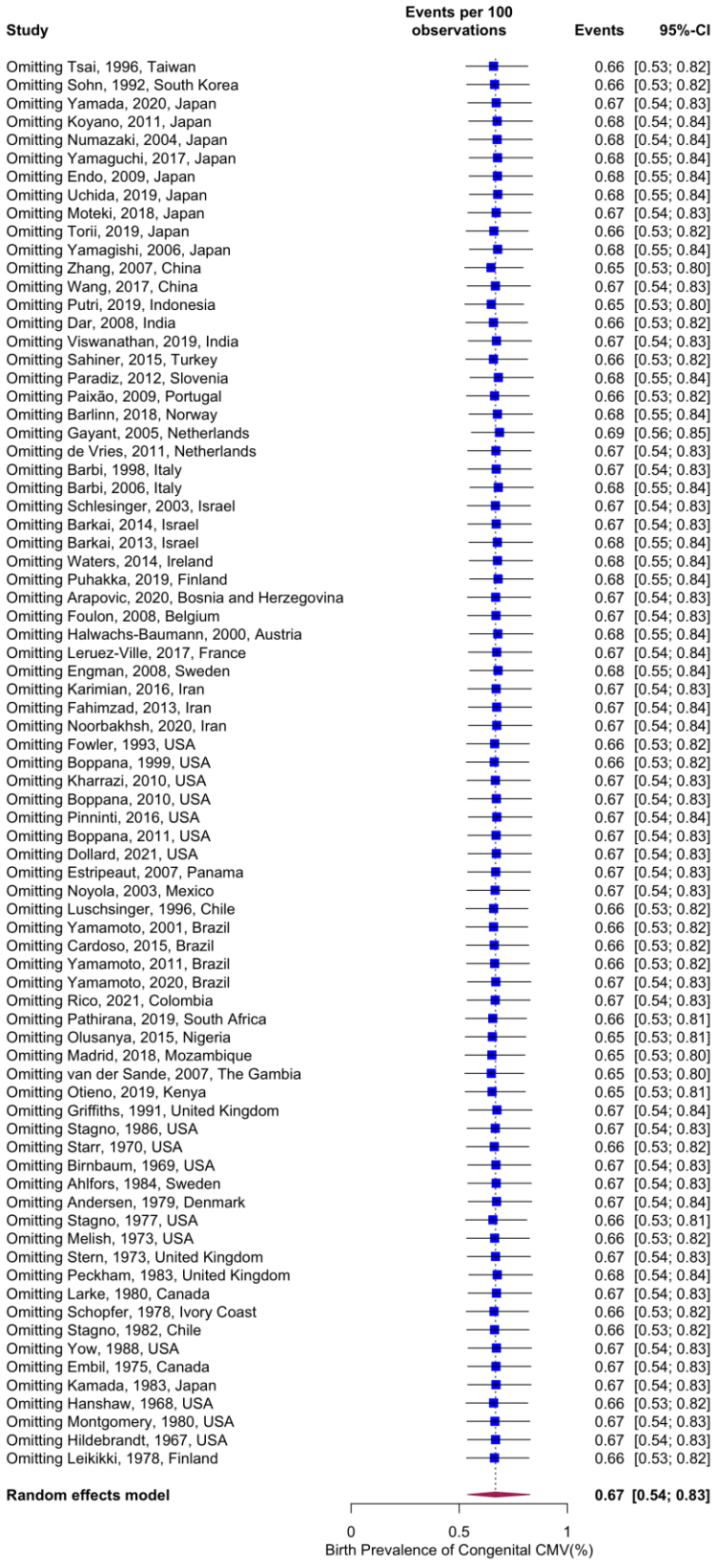

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eFigure 5: Influence and outlier (leave-one-out meta-analysis) analysis for the birth prevalence of cCMV.[1] The results of our outlier and influence analysis show the recalculated pooled point estimate ranged from 0.65% to 0.69% when one study omitted each time.

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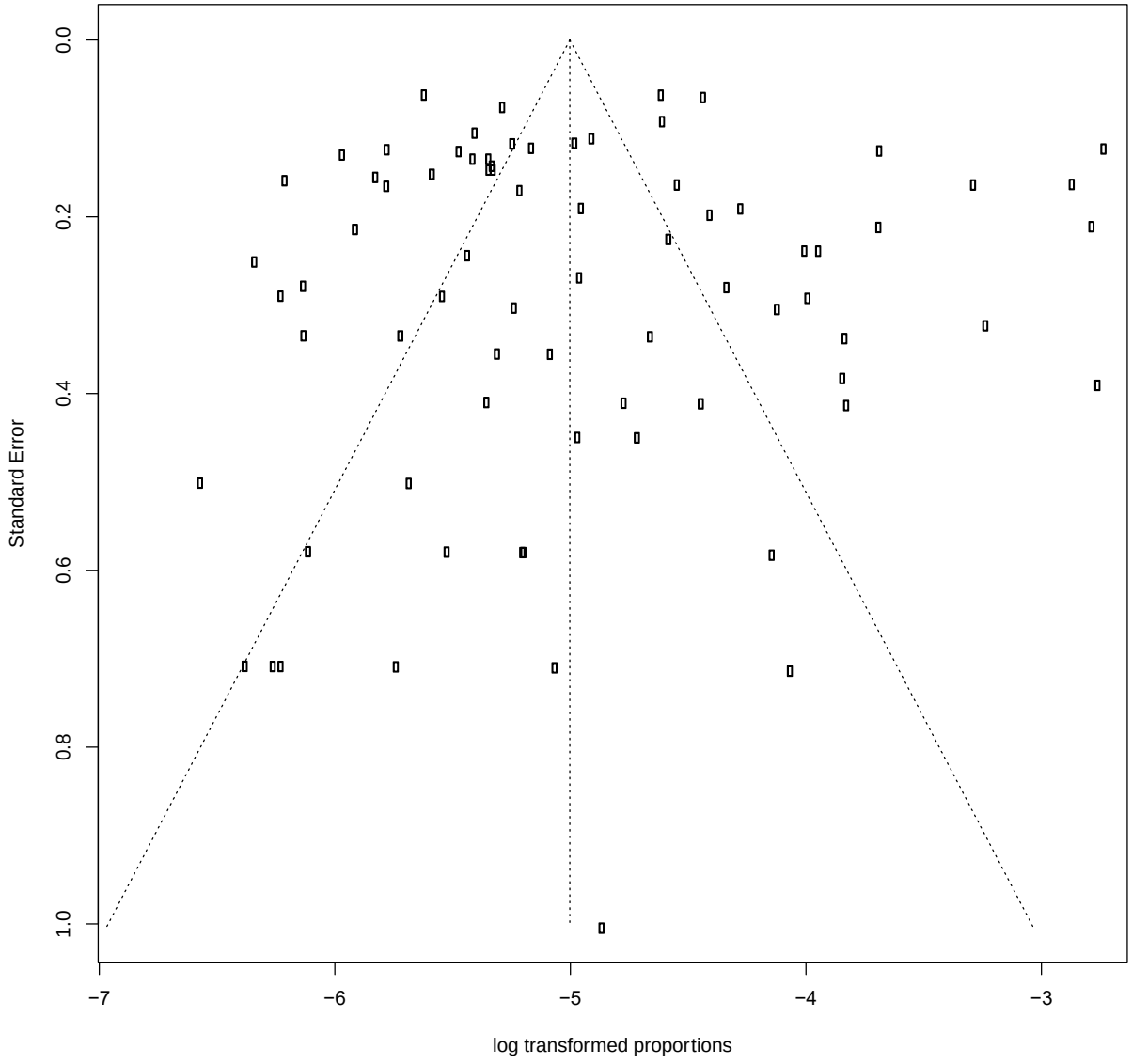

eFigure 6: Funnel plots to assess potential for small-study publication bias.[2] Symmetrical inverted funnel plot suggested absence of publication bias.

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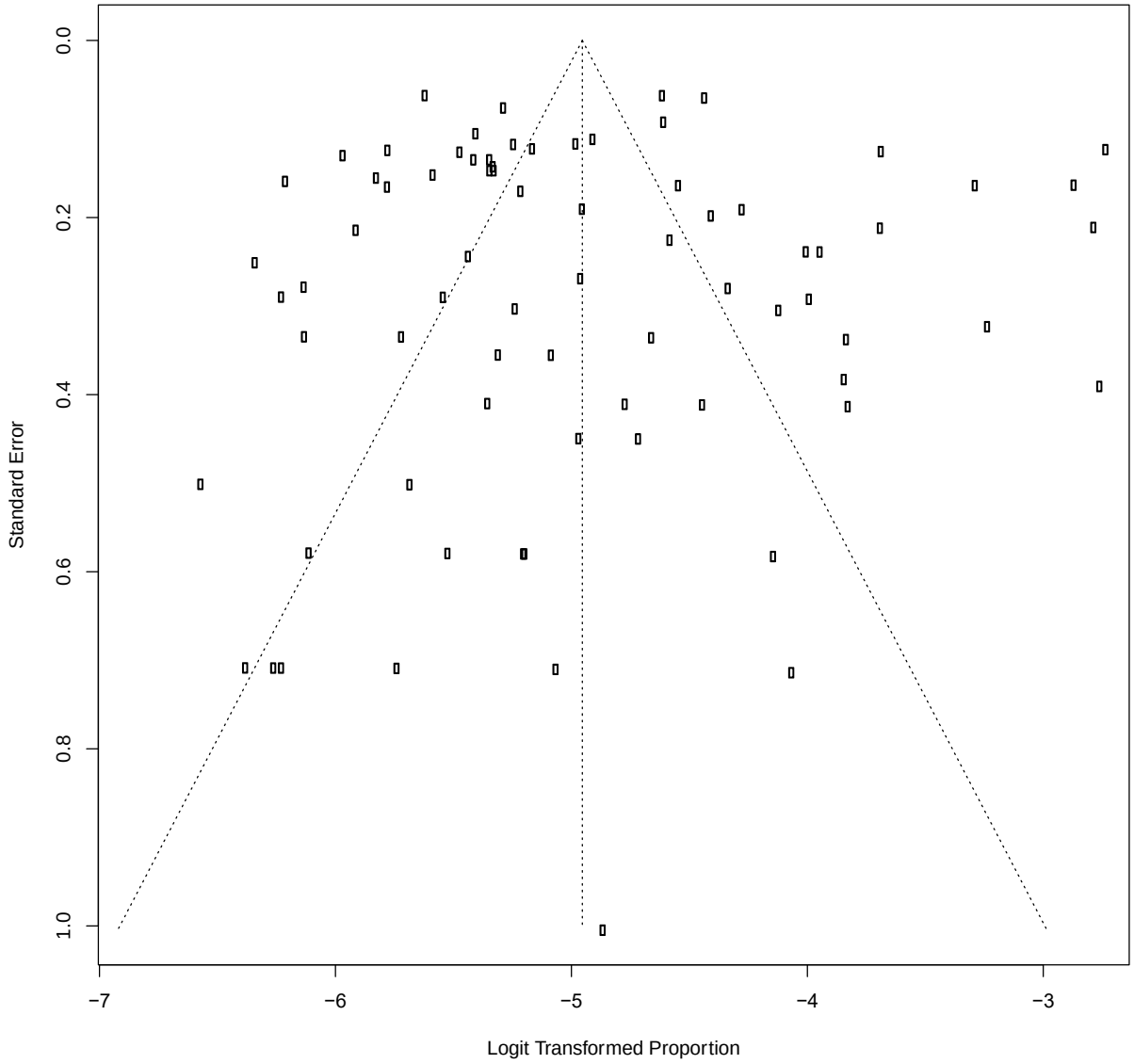

eFigure 7: Funnel plots from trim and fill analysis. Duval & Tweedie trim and fill analytical method suggests that the adjusted effect estimates would fall in the range of 0.57% to 0.87%, and no additional studies were added.[3]

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